DESIGN INTELLIGENCE · REVERSE-ENGINEERING REPORT

Vibrant.Design Teardown

A measured dissection of a production landing page: every design token, all seven easing curves in its motion library, the complete animated-check micro-interaction specification, and the section grammar that front-loads proof - extracted from live HTML and compiled CSS, then converted into recreation recipes and a 40-item agent compliance checklist.

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1. Executive Summary

This report is the written record of a reverse-engineering pass over **vibrant.design**, a design-inspiration platform that curates landing pages, sections, launch videos and UI shots from across the internet. The goal of the exercise was not to copy the site, but to understand precisely how a top-tier, motion-heavy landing page is engineered: which design tokens it commits to, how its animation system is budgeted, and which repeatable techniques produce the polish that makes the page feel expensive. Every number quoted in this document is a measured value extracted from the site's live production HTML and its compiled stylesheet, not an estimate or a recollection.

Three findings stand out. First, the visual system is radically restrained: one typeface (Inter), one saturated brand hue in two blue steps (#3074FE for interactive elements and #1838f8 for deep accents), and a neutral ramp tokenized in OKLCH. Everything else is whitespace discipline. Second, the motion system is built like a software library: a small set of named easing curves (seven distinct cubic-bezier values were recovered), strict duration ranges per interaction type, and a hard constraint that only transform, opacity, filter and background-position are ever animated. Third, the site invests its animation budget where the user is already looking: card grids, success states and ambient loops, while navigation and layout shifts stay almost instant.

38	7	1
named @keyframes recovered from production CSS	distinct easing curves in the motion library	typeface family across the entire page

The final chapter converts these findings into an actionable artifact: a compliance checklist (gates A through I) and a ready-to-run agent prompt that instructs an AI coding agent to audit a landing page against the measured benchmark and repair every failure. The prompt, together with the machine-readable token data behind this report, ships alongside this document in the accompanying repository.

2. Methodology and Data Provenance

The teardown followed a black-box, evidence-first protocol. No source maps, private repositories or build artifacts were used; every claim is grounded in what any visitor's browser downloads when it opens the page. This matters because it makes the findings reproducible: anyone with curl and a CSS beautifier can repeat the extraction and arrive at the same numbers. It also frames the limitations honestly - behaviors that live only in minified JavaScript bundles (scroll-linked physics, spring solvers) are described from observed behavior rather than decompiled source.

Evidence	Size	What it yielded
Landing page HTML (server-rendered)	247 KB	Meta tags, section order, headline and CTA copy, inline color usage, font stack declaration
Compiled stylesheet globals-7PsNk93W.css	357 KB	All design tokens, 7 easing curves, 38 keyframes, radius and blur scales, dark-theme ramp
Rendered page observation	-	Hover and entrance behavior, stagger rhythm, frosted-nav transition, marquee and loop pacing

Table 2-1: The evidence corpus behind every claim in this report.

Extraction was mechanical rather than interpretive. Design tokens were pulled by scanning the stylesheet for custom property declarations; easing curves were harvested from every transition and animation declaration and deduplicated by frequency; keyframe names were inventoried with their bodies intact so each loop could be replayed offline. Copy lengths (headline word counts, meta description character counts) were measured directly from the DOM. Where this report recommends a value that was not directly measurable - for example, the 40 to 80 millisecond stagger window - the recommendation is labeled as a synthesis from repeated observation, not a hard extracted fact.

One scope note: the site's media cards (curated videos, GIFs and UI shots) are user-facing content rather than part of the design system, so they were catalogued but not dissected frame by frame. The engineering value of this teardown sits in the system underneath that content - the tokens, curves and components - which is what a team can legitimately re-apply to its own landing page.

3. The Visual System: Color, Typography, Geometry

The visual layer is a study in constraint. A single saturated hue carries the entire brand; a strictly derived neutral ramp does everything else; one font family renders every glyph on the page; and a compact geometric scale governs corners, blur and media ratios. The subsections below give the measured values and the design logic that makes them cohere.

3.1 Color: one brand hue over an OKLCH neutral ramp

The page canvas is a warm near-white, #f7f7f7, used consistently 25 times across inline styles, with pure white card surfaces floating on it. The brand blues are few and deliberate: #3074FE is the primary interactive blue, #1838f8 a deeper electric accent, and #2563eb appears as component fill (for example, the animated checkmark studied in Chapter 5). Crucially, the blue appears only where action or aliveness is intended: primary buttons, links, live indicators. The remaining system is tokenized in OKLCH in the shadcn convention, which makes both themes mathematically symmetric rather than hand-tuned.

Role	Light theme	Dark theme	Usage discipline
Page background	`#f7f7f7`	`oklch(14.5% 0 0)` (~`#0a0a0a`)	Canvas only; never for text
Card surface	`#ffffff`	`oklch(20.5% 0 0)` (~`#171717`)	Elevation layer for media and panels
Foreground	`oklch(14.5% 0 0)`	`oklch(98.5% 0 0)`	Body and headings
Border	`oklch(92.2% 0 0)`	`oklch(100% 0 0 / .1)`	Hairlines; 10% white alpha in dark
Brand primary	`#3074FE`	`same`	Primary CTA fill, links, active states
Brand deep	`#1838f8`	`same`	Electric accent, small areas only
Component fill	`#2563eb`	`same`	Micro-component fills (check, toggles)

Table 3-1: Measured color tokens. The dark theme mirrors the light theme in the same hue space.

3.2 Typography: a single family, tightly tracked

Exactly one family - Inter, wired through the --font-inter token with a system fallback chain - renders everything from the 21-pixel headline down to 10-pixel metadata. Headlines at display sizes carry negative tracking of roughly -0.02 to -0.04 em, which is what gives large Inter its compact, product-grade density; below about 20 pixels tracking returns to normal. The measured hero copy is as disciplined as the type: an eight-word headline (Design inspiration from the best of the internet) over a nine-word sub-headline that names exactly four concrete content types. Nothing on the page is set in a second family, and no decorative weights appear anywhere, which keeps the rendering pipeline and the font payload minimal.

3.3 Geometry: radius, blur and rhythm

Corners follow a derived scale rooted at 10 pixels (0.625rem): md at 8, lg at 10, xl at 14 and 2xl at 18 (1.8 times the base). Blur is similarly stepped at 4, 8, 12 and 24 pixels and powers the frosted navigation bar and overlay surfaces, always paired with a translucent hairline border. Media cards are locked to 16:9 so masonry rows align without layout shift, and vertical rhythm runs on a 4-pixel base grid with section spacing in the 96 to 128 pixel band on desktop. None of these numbers is exotic; the refinement comes from never violating them.

Token	Measured value	Observed application
`--radius (base)`	`0.625rem = 10px`	lg corners on cards and inputs
`--radius-xl / 2xl`	`14px / 18px`	Hero panels, large media tiles
`--blur scale`	`4 / 8 / 12 / 24px`	Frosted nav (sm-md), overlays (xl)
`Aspect ratio`	`16 / 9 (--aspect-video)`	Every media card, no exceptions
`Base spacing`	`4px grid (Tailwind spacing)`	All paddings and gaps
`Section rhythm`	`96-128px vertical (desktop)`	Chapter-like separation of rails

Table 3-2: Geometry tokens and where they surface in the layout.

4. The Motion System

Motion is where vibrant.design separates itself from ordinary landing pages, and the compiled CSS proves the separation is systematic rather than accidental. Seven distinct cubic-bezier curves recur across transitions and keyframes, each with a consistent job. Durations cluster into narrow bands by interaction class: micro-feedback at 150 to 250 milliseconds, entrances at 300 to 600 milliseconds, and ambient loops at two seconds or more. The system also obeys a performance constitution: every animated property observed in the stylesheet is one of transform, opacity, filter or background-position - never width, height, top or left.

Curve	Cubic-bezier	Character	Where it is used
`ease-out-quint`	`( .22, 1, .36, 1 )`	Fast start, long glide	Default entrances, drawers, most UI motion
`ease-out-quart`	`( .23, 1, .32, 1 )`	Slightly firmer glide	Panels, cards, hover lifts
`ease-out-expo`	`( .16, 1, .3, 1 )`	Explosive settle	Hero reveals, large surfaces
`ease-in-out-quint`	`( .32, .72, 0, 1 )`	iOS-style symmetric	State changes, layout shifts
`ease-in-out-cubic`	`( .66, 0, .34, 1 )`	Neutral swap	Position exchanges, loops
`spring overshoot`	`( .34, 1.96, .94, 1 )`	Bouncy pop past 1	Success states, badge bounce
`soft overshoot`	`( .34, 1.45, .64, 1 )`	Gentle over-travel	Buttons pressed and released

Table 4-1: The seven easing curves recovered from production CSS, with the job each one does.

180ms	60ms	500ms
typical hover transition duration	observed stagger step between grid items	typical entrance duration with ease-out-expo

4.1 Keyframe inventory

Thirty-eight named keyframe animations were recovered, and they fall into four families. Skeleton and shimmer loops use background-position sweeps; ambient loops breathe opacity and scale; entrance and exit animations compose translation, scale and rotation variables in the tw-animate style; and utility loops (ping, pulse, spin) cover status indicators. The naming alone - streaming-word-fade, digit-pop-in, otp-bounce, scan-line - shows a team that treats animation as an audited surface, not a garnish.

Keyframe	Pattern (measured)	Class
`shimmer`	background-position 200% 0 to -200% 0	Skeleton / text sheen
`skeleton-shimmer / tw-shimmer`	Same sweep, surface variants	Loading placeholders
`orb-wave`	opacity rest, 1 @ 28%; scale 1, 1.18, 1	Ambient breathing orbs
`enter / exit`	translate3d + scale3d + rotate from vars	Entrance system
`text-shimmer / streaming-word-fade`	Position sweep / per-word opacity	AI-style streaming text
`digit-pop-in / otp-bounce`	Small translateY + scale pops	Numeric feedback
`scan / scan-line`	Linear sweep with fading tail	Scanning states
`ping / pulse / spin`	1s / 2s / 1s standard loops	Status indicators

Table 4-2: Keyframe families. Periods are measured; all loops pause under prefers-reduced-motion.

4.2 Motion laws worth copying

- **Entrances**: 300-600ms, travel distance of 24px or less, staggered 40-80ms per item, capped at roughly eight staggered children before the rest appear together.
- **Hovers**: 150-250ms, scale no greater than 1.03, expressed as shadow or opacity shifts so layout never reflows.
- **Ambient loops**: period of 2s or longer, opacity or scale excursions bounded near 1.18, always paused under reduced-motion preferences.
- **Property whitelist**: transform, opacity, filter, background-position. Animating layout properties is treated as a defect, not a style choice.

- **One system, many surfaces**: the same seven curves appear in the nav, the cards, the pricing checks and the footer newsletter interaction, which is why the page feels coherent.

5. Micro-Interaction Anatomy: The Animated Check

One component repays close study: the animated checkmark used in pricing and task states, whose full specification survives in the stylesheet as a set of custom properties. It is a masterclass in sequencing - a three-act animation (draw, fill, pop) with measured durations, a progress lead, and an overshoot curve - in roughly twenty lines of CSS. The table below is the complete recovered spec; recreating it verbatim takes about ten minutes and instantly upgrades any checklist UI.

Act	Duration / curve	Measured parameters
Arc draw	`0.3s, ease-out-quint ( .22, 1, .36, 1 )`	stroke draw with 0.64 progress lead
Fill wipe	`0.22s, ( .34, 1.35, .64, 1 )`	fill color #2563eb sweeps the disc
Success pop	`0.22s, spring ( .34, 1.96, .94, 1 )`	scale to 1.09, then settle
Revert	`0.2s, ease-out-quint`	returns to outline state
Ring	`static`	22px size; ring width = size x 2.5 / 22; rgba(0,0,0,.10) light, rgba(255,255,255,.12) dark
Blur	`static`	0.5px soft edge on the ring for anti-aliasing warmth

Table 5-1: Complete measured specification of the animated check component.

Two details deserve emphasis. The 0.64 lead means the arc begins its fill before it finishes drawing, so the acts overlap and the eye reads one continuous gesture rather than three steps. And the pop curve peaks at 1.96 on its y2 control point - an aggressive overshoot - but is tamed by the tiny 1.09 target scale, producing bounce that reads as confidence rather than cartoon. This is the level of intentionality the rest of the report's recommendations aim at.

> *"Sequencing beats spectacle: three short acts with one shared easing family outperform any single long animation."*

6. Page Architecture and Copy Engineering

The landing page is assembled from a fixed section grammar, delivered in an order that front-loads proof. Rather than describing the product with adjectives, the hero hands the visitor the product itself - a live, scrolling grid of named curated cards - within one screen. SEO metadata is similarly engineered: the title weighs 38 characters and the description 97, both comfortably inside search truncation limits.

Order	Section	Measured spec and role
1	Announcement strip	40px pill row, rotating Latest tags, dismissible, zero CLS
2	Sticky nav	64px; frosted (8-12px blur) after 8px of scroll; hairline border
3	Hero	8-word H1, 9-word sub, CTA pair: Surprise me + Explore Moodboards
4	Live product grid	16:9 masonry of named cards; content doubles as social proof
5	Sponsored slot	Single labeled placement between grid rows
6	Category rails	Hero, Features, Bento, Pricing, CTA, Logo Cloud, Testimonials, FAQ, Changelog
7	Utility pages	How it works, Integrations, Stats, Careers, Blogs, Docs, Login/Signup, 404
8	Footer	Link columns plus oversized wordmark as closing anchor

Table 6-1: The measured section grammar, top to bottom.

6.1 Copy formulas extracted from the page

- **Headline**: benefit-led, at most ten words, zero abstractions. The measured H1 names the audience payoff and the source in eight words.
- **Sub-headline**: one line, at most twenty words, enumerating concrete deliverables - the measured version lists exactly four: websites, sections, launch videos, UI shots.
- **CTAs**: verb-first with a point of view. Surprise me converts exploration into a game; Explore Moodboards names the destination. No Submit or Learn more anywhere.
- **Proof placement**: the grid itself is the testimonial. Names like Melting Collaborate Button and Unfolding Crumpled Paper Effect do the persuading that stock logos usually fake.
- **Metadata**: title 38 characters; description 97 characters; both under truncation limits while carrying the four content-type keywords.

7. Recreation Recipes

This chapter translates the extracted specs into four self-contained recipes a developer can paste into any codebase. Each recipe uses only the measured values from Chapters 3 through 5 and respects the property whitelist, so every example runs at compositor speed and degrades gracefully under reduced-motion preferences.

7.1 Skeleton shimmer

The shimmer is a two-stop background sweep on a neutral gradient. Keep the loop at 1.5 seconds or slower; anything faster reads as an alarm rather than patience.

shimmer.css

```
.skeleton {
background: linear-gradient(90deg, #eef3fa 25%, #e4ecf5 37%, #eef3fa 63%);
background-size: 400% 100%;
animation: shimmer 1.6s cubic-bezier(.22, 1, .36, 1) infinite;
}
@keyframes shimmer {
0% { background-position: 200% 0; }
to { background-position: -200% 0; }
}
```

7.2 Ambient breathing orbs

Ambient depth in the final call-to-action comes from overlapping radial fields that breathe on a long period. Opacity excursions stay subtle and the loop is trivially pausable, satisfying the ambient-loop law from Chapter 4.

orb-wave.css

```
.orb { animation: orb-wave 2.4s cubic-bezier(.66, 0, .34, 1) infinite; }
@keyframes orb-wave {
0% { opacity: var(--orb-rest, .4); scale: 1; }
28% { opacity: 1; scale: 1.18; }
56%, to { opacity: var(--orb-rest, .4); scale: 1; }
}
@media (prefers-reduced-motion: reduce) { .orb { animation: none; } }
```

7.3 Staggered reveal utility

One IntersectionObserver plus a CSS custom property produces the entire entrance system. The index variable sets the stagger step; the observer un-watches after reveal so scrolling stays cheap.

reveal.css + one observer

```
.reveal { opacity: 0; transform: translateY(16px); }
.reveal.is-in { opacity: 1; transform: none;
transition: opacity .5s cubic-bezier(.16, 1, .3, 1),
transform .5s cubic-bezier(.16, 1, .3, 1);
transition-delay: calc(var(--i, 0) * 60ms); }
// js: io = new IntersectionObserver(es => es.forEach(e => {
// if (e.isIntersecting) { e.target.classList.add("is-in"); io.unobserve(e.target);
}
// }), { threshold: .15 });
```

7.4 The success pop

Finally, the check component's last act as a one-liner pattern: whenever a state turns successful, pop to 1.09 with the aggressive spring curve, then let it settle. It is the cheapest delight-per-line ratio in the entire system.

success-pop.css

```
.pop-in { animation: pop .22s cubic-bezier(.34, 1.96, .94, 1); }
@keyframes pop { 0% { scale: .8; } to { scale: 1.09; } }
/* settle: follow with a .18s ease-out-quint transition back to scale: 1 */
```

8. Application: Checklist Integration and Recommendations

The teardown ends where the previous deliverable began: the agent prompt. The measured values in this report are the quantitative backbone of a 40-item compliance checklist (gates A through I) covering performance, accessibility, SEO, responsive behavior, content quality, visual system, motion, conversion and engineering hygiene. That checklist, plus the exact token values, easing curves and section grammar measured here, ships as a single self-contained prompt file designed to be pasted into an AI coding agent such as Cursor or Claude Code.

Used together, the report and the prompt close the loop in both directions. An agent fixing a landing page can cite the benchmark for every decision - why the radius scale bottoms out at 10 pixels, why hovers finish inside 250 milliseconds, why only compositor-friendly properties animate - and a human reviewer can audit the agent's output against the same numbers, since every checklist gate demands pasted evidence rather than self-attestation.

8.1 Recommendations for teams adopting this system

- **Adopt the restraint first**: one typeface, one brand hue, derived neutrals. Eighty percent of the perceived quality comes from deleting options, not adding effects.

- **Stand up the motion tokens early**: ship the seven-curve library and duration bands as CSS custom properties on day one so no engineer ever invents a ninth easing.
- **Buy delight where it is cheap**: success pops, skeleton shimmer and breathing orbs cost a few dozen lines combined and carry most of the perceived polish.
- **Let content be the social proof**: name and show real artifacts instead of stock endorsement walls.
- **Audit continuously**: grep for raw hex values, transition:all and layout-property keyframes in CI; the system rots exactly as fast as its exceptions accumulate.

Reproducing these measurements requires only the evidence listed in Table 2-1: a fetch of the landing page and its compiled stylesheet. Token values, keyframe bodies and copy measurements referenced throughout this document are also provided in machine-readable form (data/extracted-tokens.json) in the companion repository, so the benchmark can be diffed against future redesigns of the source site.